P01. A Priority-Aware Replanning and Resequencing Framework for Coordination of Connected and Automated Vehicles

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1. 论文全景速览与核心贡献 / High-Level Overview & Core Contributions

研究背景与痛点 / Background & Pain Points

这篇论文位于“Scheduling/resequencing + optimal control”方向。对你的 JSSP-based Intersection Management 课题，它回答的是高层调度、低层轨迹平滑、安全过滤或真实验证中的一个关键子问题：如何让车辆在路口少停、少冲突、少能耗，同时保持在线可执行。

The paper belongs to the theme of Scheduling/resequencing + optimal control. For a JSSP-based intersection-management thesis, it illuminates one of the key layers: scheduling, smoothing, safety filtering, or real-world validation.

Assumes connected automated vehicles, known paths, and simplified dynamics; the paper is closer to resequencing theory than to learned graph scheduling.

核心科学问题 / Core Scientific Question

当车辆到达扰动使原始通行序列不再高效或可行时，如何在不牺牲安全约束的前提下快速重排序并生成可执行轨迹?

How can an intersection manager resequence disturbed arrivals while preserving safety and low-level trajectory feasibility?

科学贡献 / Scientific Contributions

- 方法贡献 (method contribution)：Event-driven replanning and priority-aware resequencing layered on decentralized CAV optimal control at a signal-free intersection.
- 安全/避碰贡献 (safety contribution)：Uses rear-end and lateral safety constraints around conflict points, with replanning when disturbances change feasibility.
- 平滑/停止避免贡献 (smoothing contribution)：The lower-level optimal-control trajectory is constructed so vehicles can conserve momentum rather than stop at the intersection when a feasible exit time exists.
- 创新力度评判 (reviewer judgement)：强相关但中等创新：把 resequencing 明确嵌入 CAV 最优控制闭环，贡献主要在系统架构和事件触发逻辑。

核心概念对照 / Core Concepts

中文概念	English Concept	Role / 学术作用
自动交叉口管理	Autonomous Intersection Management	把信号控制替换为车辆级调度与控制。
轨迹平滑	Trajectory Smoothing	减少急加减速、停车和能耗峰值。
安全约束	Safety Constraints	把碰撞风险写成距离、时间间隔或屏障函数。

Priority-aware resequencing + decentralized optimal control	优先级重排序与分散最优控制	该论文的核心方法族。

教材式学习路径 / Textbook-Style Study Path

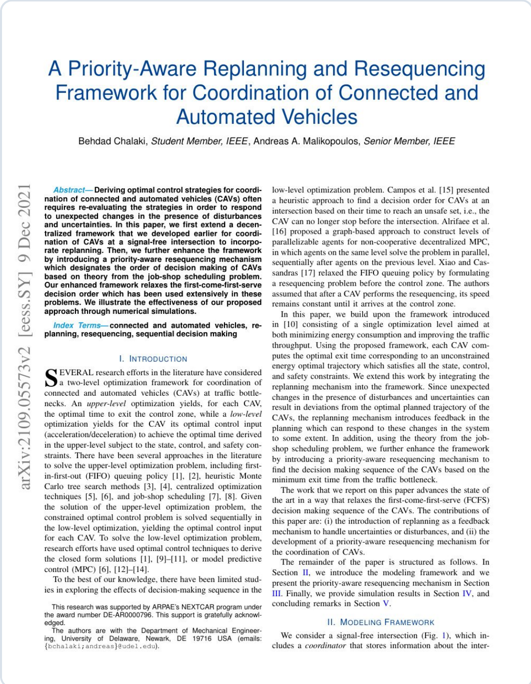
- 第一遍先读首页截图、摘要与引言，确认研究问题和场景假设。
- 第二遍沿着技术路线图复现模型变量、约束和求解器输入输出。
- 第三遍逐节阅读 section deep reading，对照 PDF 截图检查公式、图表和实验。
- 第四遍只站在审稿人视角重读实验，判断 baseline、公平性、消融和鲁棒性。
- 最后把博士 checklist 写成自己的复现实验或论文拓展计划。

博士阅读 Checklist / Doctoral Reading Checklist

- 能否独立重写核心公式：双积分车辆模型与能量型最优控制 / Double-integrator vehicle model with energy-like optimal control，并解释每个符号的物理意义？
- 能否画出输入-模型-算法-控制-验证的数据流图？
- 能否指出至少两个强假设，并说明它们在真实交通中何时失效？
- 能否复述实验基准是否公平，指标是否覆盖效率、安全、能耗和计算时间？
- 能否提出一个与自己 JSSP/RL/smoother 课题直接相连的拓展实验？

关键截图讲解 / PDF Snapshot Explanation

截图来自本地 PDF 的精选页面，用于学习定位和版面理解；请与原 PDF 平行阅读，不把截图视作全文替代。



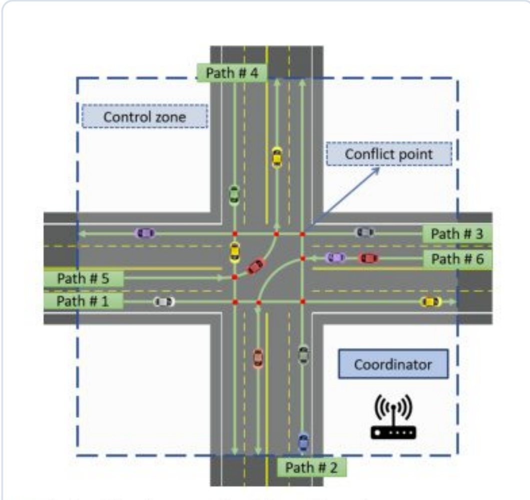
首页与摘要 / Title and abstract page

p.1 · full_page

这张截图用于把笔记中的抽象概念拉回原论文页面，便于你平行阅读原文、公式、图表和实验设置。

This snapshot anchors the study note to the original PDF page so you can read the explanation alongside the source layout.

与当前课题的连接：Directly supports this project's hierarchy: a high-level sequence or schedule provides target times, and a low-level controller checks executable motion.



g. 1. A signal free intersection with conflict points.

建模或问题设定页 / Modeling or problem-setup page

p.2 · figure_crop

这张截图用于把笔记中的抽象概念拉回原论文页面，便于你平行阅读原文、公式、图表和实验设置。

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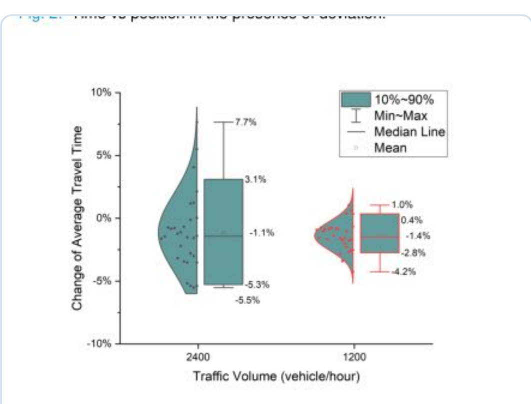

Fig. 3. Change in average travel time compared to the FCFS decision

实验或结果页 / Experiments or results page

p.5 · figure_crop

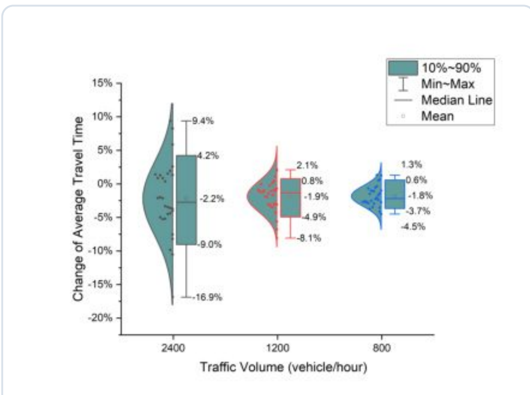

Fig. 4. Change in weighted average travel time compared to the FCFS

结论或局限页 / Conclusion or limitations page

p.6 · figure_crop

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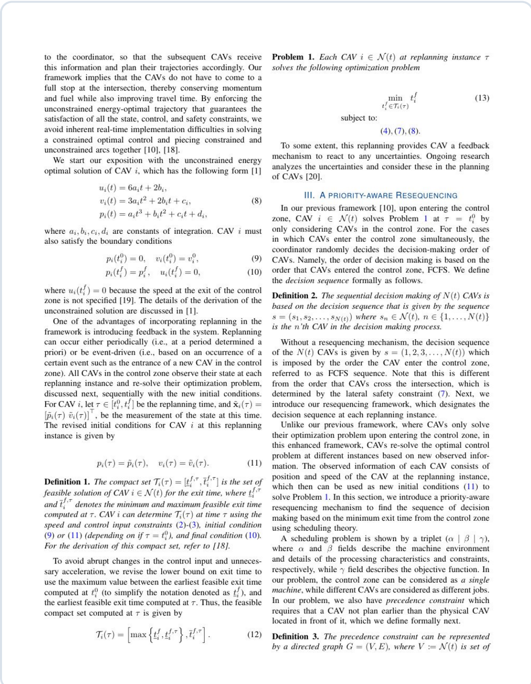

to the coordinator, so that the subsequent CAVs receive this information and plan their trajectories accordingly. Our framework implies that the CAVs do not have to come to a full stop at the intersection, thereby conserving momentum and fuel while also improving travel time. By enforcing the unconstrained energy-optimal trajectory that guarantees the satisfaction of all the state, control, and safety constraints, we avoid inherent real-time implementation difficulties in solving a constrained optimal control and piecing constrained and unconstrained arcs together [10], [18].

We start our exposition with the unconstrained energy optimal solution of CAV i, which has the following form [1]

$$\begin{aligned} u_i(t) &= 6a_i t + 2b_i, \\ v_i(t) &= 3a_i t^2 + 2b_i t + c_i, \\ p_i(t) &= a_i t^3 + b_i t^2 + c_i t + d_i, \end{aligned} \tag{8}$$

where a_i, b_i, c_i, d_i are constants of integration. CAV i must also satisfy the boundary conditions

$$p_i(t_i^0) = 0, \quad v_i(t_i^0) = v_i^0, \tag{9}$$

$$p_i(t_i^f) = p_i^f, \quad u_i(t_i^f) = 0, \tag{10}$$

where $u_i(t_i^f) = 0$ because the speed at the exit of the control zone is not specified [19]. The details of the derivation of the unconstrained solution are discussed in [1].

One of the advantages of incorporating replanning in the framework is introducing feedback in the system. Replanning can occur either periodically (i.e., at a period determined a priori) or be event-driven (i.e., based on an occurrence of a certain event such as the entrance of a new CAV in the control zone). All CAVs in the control zone observe their state at each replanning instance and re-solve their optimization problem, discussed next, sequentially with the new initial conditions. For CAV i, let $\tau \in [t_i^0, t_i^f]$ be the replanning time, and $\tilde{\mathrm{x}}_i(\tau) = [\tilde{p}_i(\tau)\ \tilde{v}_i(\tau)]^\top$, be the measurement of the state at this time. The revised initial conditions for CAV i at this replanning instance is given by

$$p_i(\tau) = \tilde{p}_i(\tau), \quad v_i(\tau) = \tilde{v}_i(\tau). \tag{11}$$

Definition 1. *The compact set $\mathcal{T}_i(\tau) = [\underline{t}_i^{f,\tau}, \bar{t}_i^{f,\tau}]$ is the set of feasible solution of CAV $i \in \mathcal{N}(t)$ for the exit time, where $\underline{t}_i^{f,\tau}$ and $\bar{t}_i^{f,\tau}$ denotes the minimum and maximum feasible exit time computed at τ. CAV i can determine $\mathcal{T}_i(\tau)$ at time τ using the speed and control input constraints (2)-(3), initial condition (9) or (11) (depending on if $\tau = t_i^0$), and final condition (10). For the derivation of this compact set, refer to [18].*

To avoid abrupt changes in the control input and unnecessary acceleration, we revise the lower bound on exit time to use the maximum value between the earliest feasible exit time computed at t_i^0 (to simplify the notation denoted as $\underline{t}_i^f$), and the earliest feasible exit time computed at τ. Thus, the feasible compact set computed at τ is given by

$$\mathcal{T}_i(\tau) = \left[\max\left\{\underline{t}_i^f, \underline{t}_i^{f,\tau}\right\}, \bar{t}_i^{f,\tau}\right]. \tag{12}$$

Problem 1. *Each CAV $i \in \mathcal{N}(t)$ at replanning instance τ solves the following optimization problem*

$$\min_{t_i^f \in \mathcal{T}_i(\tau)} t_i^f \tag{13}$$

subject to:

(4), (7), (8).

To some extent, this replanning provides CAV a feedback mechanism to react to any uncertainties. Ongoing research analyzes the uncertainties and consider these in the planning of CAVs [20].

III. A PRIORITY-AWARE RESEQUENCING

In our previous framework [10], upon entering the control zone, CAV $i \in \mathcal{N}(t)$ solves Problem 1 at $\tau = t_i^0$ by only considering CAVs in the control zone. For the cases in which CAVs enter the control zone simultaneously, the coordinator randomly decides the decision-making order of CAVs. Namely, the order of decision making is based on the order that CAVs entered the control zone, FCFS. We define the *decision sequence* formally as follows.

Definition 2. *The sequential decision making of $N(t)$ CAVs is based on the decision sequence that is given by the sequence $s = (s_1, s_2, \ldots, s_{N(t)})$ where $s_n \in \mathcal{N}(t)$, $n \in \{1, \ldots, N(t)\}$ is the n'th CAV in the decision making process.*

Without a resequencing mechanism, the decision sequence of the $N(t)$ CAVs is given by $s = (1, 2, 3, \ldots, N(t))$ which is imposed by the order the CAV enter the control zone, referred to as FCFS sequence. Note that this is different from the order that CAVs cross the intersection, which is determined by the lateral safety constraint (7). Next, we introduce our resequencing framework, which designates the decision sequence at each replanning instance.

Unlike our previous framework, where CAVs only solve their optimization problem upon entering the control zone, in this enhanced framework, CAVs re-solve the optimal control problem at different instances based on new observed information. The observed information of each CAV consists of position and speed of the CAV at the replanning instance, which then can be used as new initial conditions (11) to solve Problem 1. In this section, we introduce a priority-aware resequencing mechanism to find the sequence of decision making based on the minimum exit time from the control zone using scheduling theory.

A scheduling problem is shown by a triplet $(\alpha \mid \beta \mid \gamma)$, where α and β fields describe the machine environment and details of the processing characteristics and constraints, respectively, while γ field describes the objective function. In our problem, the control zone can be considered as *a single machine*, while different CAVs are considered as different jobs. In our problem, we also have *precedence constraint* which requires that a CAV not plan earlier than the physical CAV located in front of it, which we define formally next.

Definition 3. *The precedence constraint can be represented by a directed graph $G = (V, E)$, where $V := \mathcal{N}(t)$ is set of*

补充精读页 / Supplemental close-reading page

p.3 · full_page

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与当前课题的连接：Directly supports this project's hierarchy: a high-level sequence or schedule provides target times, and a low-level controller checks executable motion.

章节精读 / Section-by-Section Deep Reading

p.1 · motivation

I. INTRODUCTION

章节目标： 建立研究动机：为什么传统信号控制、FCFS 或单车控制不足以处理该论文关注的交叉口协同问题。

Learning goal: Understand why the paper needs a new coordination method rather than a standard signal, FCFS, or single-vehicle controller.

关键论点 / Key argument: 这一部分通常把痛点落到 Scheduling/resequencing + optimal control：既要提高效率，又要保持安全与在线可执行。

技术焦点 / Technical focus: 精读时标出作者批判的 baseline、场景假设、交通参与者类型和隐含优化目标。

审稿追问 / Reviewer question: 作者指出的痛点是否真的需要新方法，还是可以由更强的优化器/更保守规则解决？

p.1 · method

II. M ODELING FRAMEWORK

章节目标： 解释算法如何把模型转化为可执行决策，并说明在线运行机制。

Learning goal: Trace how the paper turns the model into executable online decisions.

关键论点 / Key argument: 高层监测车辆进入、延误和可行性破坏；当扰动出现时，按照优先级规则更新局部通行序列，再把目标离开时间传给每辆车的解析/数值最优控制器。低层控制器检查速度、加速度、追尾和横向冲突约束是否可执行。

技术焦点 / Technical focus: 画出输入、核心求解器、输出和安全检查接口，明确哪些步骤是优化、哪些是规则、哪些是学习。

审稿追问 / Reviewer question: 若算法某一步失败，论文有没有 fallback、重规划或安全过滤机制？

p.3 · bridge

III. A PRIORITY-AWARE RESEQUENCING

章节目标： 承接前后章节，补充局部定义、案例或实现细节。

Learning goal: Recover implementation details needed for reproduction.

关键论点 / Key argument: 这类章节常常不是主贡献，但决定你能否真正复现方法。

技术焦点 / Technical focus: 检查它是否给出了参数、伪代码、场景划分或实现细节。

审稿追问 / Reviewer question: 跳过该节会不会导致公式或实验无法复现？

p.5 · experiment

IV. SIMULATION RESULTS

章节目标： 验证方法是否在公平基准下改善效率、安全、能耗或计算时间。

Learning goal: Check whether the claimed improvements are supported by fair baselines and meaningful metrics.

关键论点 / Key argument: 实验应与论文声称的贡献闭环：Numerical simulation of signal-free intersection coordination under resequencing and replanning.

技术焦点 / Technical focus: 重点追踪指标：通行时间 (travel time)、延误 (delay)、停止次数 (number of stops)、控制能量代理 (control effort) 和安全约束违反。 同时检查仿真平台、交通需求和随机性设置。

审稿追问 / Reviewer question: 实验是否只展示平均收益，还是覆盖了高密度、异常扰动和失败案例？

p.6 · critique

V. CONCLUDING REMARKS AND DISCUSSION

章节目标：提炼作者承认的边界，并把它转化为博士课题的可拓展问题。

Learning goal: Convert acknowledged boundaries into research opportunities.

关键论点 / Key argument: 结论不只是总结结果，更是观察作者没有解决什么的窗口。

技术焦点 / Technical focus: 把 future work 与前文假设逐一对应，寻找可发表的补强点。

审稿追问 / Reviewer question: 如果我是审稿人，我会要求补哪一个实验、定理或消融?

逐段式导读 / Paragraph-Level Walkthrough

Opening motivation / 开篇动机段

先把作者的痛点翻译成一句博士问题：当车辆到达扰动使原始通行序列不再高效或可行时，如何在不牺牲安全约束的前提下快速重排序并生成可执行轨迹？ 读这一段时不要急着接受动机，要追问传统 FCFS、信号灯或保守 MPC 为什么不足。

Turn the opening motivation into one doctoral-level question and test whether the paper truly needs a new method.

Assumption and setting / 假设与场景段

把车辆类型、通信条件、路口类型和动力学假设列成表。该文的关键边界包括：所有车辆为联网自动驾驶车辆 (connected automated vehicles, CAVs)，且路径已知。；车辆动力学可由简化双积分模型 (double-integrator model) 充分描述。；通信、定位和控制执行误差较小，扰动可被及时探测。

List vehicle type, communication, intersection setting, and dynamics assumptions before reading the equations.

Model and notation / 模型与符号段

围绕核心公式“双积分车辆模型与能量型最优控制 / Double-integrator vehicle model with energy-like optimal control”复写符号表，确认每个变量是决策变量、状态变量、参数还是约束阈值。

Rewrite the symbol table around the core formulation and classify variables as states, decisions, parameters, or constraints.

Algorithm mechanism / 算法机制段

把算法读成一条数据流：输入交通状态，执行 Event-driven replanning and priority-aware resequencing layered on decentralized CAV optimal control at a signal-free intersection.，输出可执行通行/控制决策，并检查安全接口。

Read the algorithm as a dataflow from traffic state to executable sequence/control decision, with explicit safety interfaces.

Experiment and evidence / 实验证据段

不要只看作者报告的提升，要检查基准、公平性和指标。本文验证描述为：Numerical simulation of signal-free intersection coordination under resequencing and replanning.

Go beyond reported gains and inspect baselines, fairness, metrics, and computation time.

Reviewer synthesis / 审稿式总结段

最后把贡献拆成可发表与需补强两部分。对你课题最直接的用途是：Directly supports this project's hierarchy: a high-level sequence or schedule provides target times, and a low-level controller checks executable motion.

Separate publishable contribution from missing evidence, then connect the result to your own thesis architecture.

技术路线图与贡献量评估 / Technical Route & Contribution Matrix

1. 输入
Input

车辆状态、路径/车道、交通需求、信号或协同信息。

2. 建模
Modeling

双积分车辆模型与能量型最优控制 / Double-integrator vehicle model with energy-like optimal control

3. 核心求解
Core solver

Event-driven replanning and priority-aware resequencing layered on decentralized CAV optimal control at a signal-free intersection.

4. 安全接口
Safety interface

Uses rear-end and lateral safety constraints around conflict points, with replanning when disturbances change feasibility.

5. 平滑与效率
Smoothing and efficiency

The lower-level optimal-control trajectory is constructed so vehicles can conserve momentum rather than stop at the intersection when a feasible exit time exists.

6. 验证
Validation

Numerical simulation of signal-free intersection coordination under resequencing and replanning.

贡献量评估 / Contribution Strength Matrix

Dimension / 维度	Score / 评分	Comment / 评语
理论贡献 / Theoretical contribution	3/5	双积分车辆模型与能量型最优控制 / Double-integrator vehicle model with energy-like optimal control
算法贡献 / Algorithmic contribution	5/5	Event-driven replanning and priority-aware resequencing layered on decentralized CAV optimal control at a signal-free intersection.
实验贡献 / Experimental contribution	3/5	Numerical simulation of signal-free intersection coordination under resequencing and replanning.
工程贡献 / Engineering contribution	4/5	Uses rear-end and lateral safety constraints around conflict points, with replanning when disturbances change feasibility.
博士课题价值 / PhD relevance	5/5	Directly supports this project's hierarchy: a high-level sequence or schedule provides target times, and a low-level controller checks executable motion.

2. 理论方法论深度解构 / Deep Dive into Methodology & Theoretical Framework

问题数学建模 / Mathematical Formulation

双积分车辆模型与能量型最优控制 / Double-integrator vehicle model with energy-like optimal control

$$
\min_{u_i(\cdot),\,t_i^f}\; J_i=\frac{1}{2}\int_{t_i^0}^{t_i^f}u_i(t)^2\,dt,\quad \dot p_i(t)=v_i(t),\quad \dot v_i(t)=u_i(t)
$$

Symbol / 符号	含义	Meaning	Role / 建模作用
i	车辆索引	vehicle index	把多车协调拆成单车最优控制子问题
$p_i(t)$	沿路径位置	path-coordinate position	描述车辆是否接近/离开冲突区
$v_i(t)$	速度	speed	连接安全间距、通行时间与停止避免
$u_i(t)$	加速度控制	acceleration control	优化变量，平滑性和能耗代理
t_i^f	车辆离开控制区时间	terminal exit time	高层排序传给低层控制器的关键接口

核心算法架构 / Core Algorithm Layout

高层监测车辆进入、延误和可行性破坏；当扰动出现时，按照优先级规则更新局部通行序列，再把目标离开时间传给每辆车的解析/数值最优控制器。低层控制器检查速度、加速度、追尾和横向冲突约束是否可执行。

The upper layer detects event-triggered infeasibility, resequences vehicles under priority rules, and sends target exit times to decentralized optimal controllers that enforce motion and safety constraints.

收敛性、稳定性或实时性 / Convergence, Stability, Real-Time Feasibility

实时性来自局部事件触发与低维车辆模型；安全性依赖硬约束而非学习策略。它不证明全局最优，但把全局重排问题约束在小窗口内，适合在线运行。

Real-time behavior comes from event-triggered local resequencing and low-dimensional dynamics; safety is constraint-based, while global optimality is intentionally relaxed.

潜在假设与边界条件 / Assumptions & Boundary Conditions

- 所有车辆为联网自动驾驶车辆（connected automated vehicles, CAVs），且路径已知。
- 车辆动力学可由简化双积分模型（double-integrator model）充分描述。
- 通信、定位和控制执行误差较小，扰动可被及时探测。

3. 实验验证与结果评判 / Experimental Validation & Result Evaluation

基准对比与环境 / Baselines & Environment

通常与固定先来先服务/不重排序 (FCFS / no resequencing) 策略比较，重点观察扰动后恢复效率。

Numerical simulation of signal-free intersection coordination under resequencing and replanning.

核心指标 / Metrics

通行时间 (travel time)、延误 (delay)、停止次数 (number of stops)、控制能量代理 (control effort) 和安全约束违反。

消融实验 / Ablation

论文更像机制验证而非完整消融；最应补充的是关闭优先级、关闭事件触发和扩大重排窗口的对比。

博士阅读提醒 / Reading Reminder

阅读实验时不要只看平均值；应检查交通密度、车流方向、随机种子、失败案例和计算耗时。For doctoral reading, inspect density regimes, demand patterns, random seeds, failure cases, and computation time rather than only mean improvements.

图表阅读线索 / Figure and Table Reading Cues

PDF extraction detected approximately 11 figure references and 0 table references. Exact captions remain in the original PDF; the bilingual cues below tell you what to inspect first.

图表名称	Figure/Table Name	Reading focus / 阅读重点
系统框架图	system framework diagram	先确认输入、决策层、控制层和安全约束之间的数据流。
冲突关系图 / 通行顺序图	conflict-relation or passing-order graph	把节点、边类型和先后约束翻译成 JSSP 的作业、机器和析取约束。
实验指标对比图表	experimental metric comparison plots/tables	重点看平均值之外的密度变化、失败场景、计算时间和方差。

4. 审稿人视角：批判性思考与未来方向 / Critical Thinking & Future Directions

论文局限性 / Limitations & Weaknesses

- 对全 CAV 环境依赖强，混合交通中人类驾驶车辆 (HDVs) 会破坏可行性假设。
- 优先级规则可能引入公平性问题，长期低优先级车流需要饥饿避免机制。
- 简化动力学忽略执行器延迟、轮胎约束和感知误差累积。

博士课题拓展点 / Research Extensions for PhD

- 把固定优先级扩展为学习型优先级函数 (learned priority function)，并加入公平性正则。
- 将重排序窗口映射为 JSSP machine queue，用图神经网络 (GNN) 预测候选交换收益。
- 在低层加入鲁棒 MPC 或 CBF 安全过滤器，处理通信延迟和轨迹预测误差。

如何服务当前课题 / Use in This Project

Directly supports this project's hierarchy: a high-level sequence or schedule provides target times, and a low-level controller checks executable motion.

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5. 交互式可视化与 PDF 排版蓝图 / Interactive Visualization & PDF Layout Blueprint

动态参数探索 / Parameter Exploration

重排序窗口大小 / Resequencing window size: veh

5

窗口增大通常降低延误但提高计算量，并可能增加局部公平性风险。

A larger window usually reduces delay but increases computation and fairness risk.

HTML 结构化布局建议 / HTML Structuring

- 顶部固定 metadata bar：题名、作者、年份、主题、PDF 页数。
- 左侧目录/section navigator，右侧为五大精读模块。
- 公式卡片使用 <pre> 或 LaTeX block，紧跟符号表。
- 审稿人批注用 reviewer-note block，博士拓展用 research-idea list。
- 交互参数用 input[type=range] + canvas 曲线，打印 PDF 时隐藏交互控件但保留解释。

来源锚点与逐节阅读路径 / Source Anchors & Section-by-Section Reading Map

以下锚点来自本地 PDF 抽取，用于回到原文复核；笔记不保存论文全文。

Page	Extracted heading
p.1	I. INTRODUCTION
p.1	II. M ODELING FRAMEWORK
p.5	IV. SIMULATION RESULTS

p.1 · I. INTRODUCTION

定位问题背景、研究缺口和为什么现有保守/非协同方法不足。

Frames the motivation, research gap, and limits of conservative or non-cooperative methods.

p.1 · II. M ODELING FRAMEWORK

给出算法流程和求解机制，应重点追踪输入、输出和安全接口。

Gives the algorithmic mechanism; track inputs, outputs, and safety interfaces.

p.3 · III. A PRIORITY-AWARE RESEQUENCING

作为局部论证段落阅读，关注它如何服务主问题。

Read as a local argument and ask how it supports the main question.

p.5 · IV. SIMULATION RESULTS

检验方法是否真的改善效率、安全或能耗；要关注基准是否公平。

Tests efficiency, safety, or energy claims; inspect whether baselines are fair.

p.6 · V. CONCLUDING REMARKS AND DISCUSSION

总结贡献和未来方向，也常暴露作者承认的边界条件。

Summarizes contributions and often reveals author-recognized boundaries.